

Final Report: Newton–Schulz Schedule Design in Muon for Transformer Pretraining

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1 Problem Setup

Problem statement. Muon is a recently proposed optimizer that updates matrix parameters by orthogonalizing a momentum-like update direction through a short Newton–Schulz (NS) iteration [2, 3]. Recent theory has also started to analyze the convergence properties of Muon-style NS updates [4]. In practice, the iteration is controlled by quintic coefficients, so schedule design becomes a spectral design problem rather than a minor implementation detail. This project asks: *under a fixed transformer pretraining stack, how do different NS coefficient schedules change the spectral geometry of Muon updates, and how does this spectral geometric change affect validation loss and training cost?*

Mathematical formulation. For a weight matrix $W \in \mathbb{R}^{m \times n}$, AdamW treats the gradient elementwise: each entry is scaled by coordinatewise first- and second-moment statistics. Muon instead treats hidden-layer weights as matrices. Its matrix update is motivated by the polar factor of a momentum-like direction, whose nonzero singular values are all equal to one. For a rectangular matrix X , the corresponding target is semi-orthogonality:

$$X^\top X \approx I_n \quad \text{if } m \geq n, \quad XX^\top \approx I_m \quad \text{if } m < n. \quad (1)$$

Equivalently, if $X = U\Sigma V^\top$, the ideal semi-orthogonal factor is $UV^\top$.

At a high level, the matrix part of Muon can be summarized as in Algorithm 1. Non-matrix parameters such as embeddings and normalization parameters are still updated by Adam-style rules, so the defining difference from AdamW lies in how matrix-valued update directions are processed.

The reason NS iterations are relevant is that they act directly on singular values. After a normalization step, Muon repeatedly applies a quintic matrix polynomial. If X_k is the matrix after the k -th NS iteration, then

$$X_{k+1} = a_k X_k + b_k X_k X_k^\top X_k + c_k (X_k X_k^\top)^2 X_k, \quad (2)$$

or the aspect-ratio-swapped equivalent when the matrix is tall. Since this map preserves the singular vectors and transforms only the singular values, the scalar dynamics are

$$\sigma_{k+1} = p_k(\sigma_k), \quad p_k(\sigma) = a_k \sigma + b_k \sigma^3 + c_k \sigma^5. \quad (3)$$

Thus a schedule $\{(a_k, b_k, c_k)\}_{k=1}^T$ defines a composed spectral map p_T . This is the mathematical object studied in the project: different Muon variants correspond to different ways of moving singular values toward the polar-factor target.

We compare AdamW and four Muon schedule families:

- **stable5**: five stable NS steps (2.0, −1.5, 0.5);
- **fast5**: five aggressive Muon steps (3.4445, −4.7750, 2.0315);
- **manual**: fast-to-stable schedules, e.g. `manual.T5_f3_s2` means three fast steps followed by two stable steps;
- **Polar Express**: quintic coefficients generated from matrix-sign theory under a lower-bound assumption on the singular values [1].

2 Technical Approach

Experimental logic. The experiment is designed to isolate the effect of the coefficient schedule. We fix the architecture, data source, token budget, learning-rate schedule, batch structure, and all Muon hyperparameters other than the orthogonalization schedule. AdamW is included as an external baseline because it does not perform NS orthogonalization on matrix parameters. Within the Muon family, the treatment variable is the schedule-defined spectral map.

All experiments are run on one RTX 4090 GPU in BF16 precision using FineWeb-10B tokenized shards [5]. The model is an 11-layer pre-norm Transformer with model dimension 768, 6 attention heads, head dimension 128, MLP ratio 4, and sequence length 2048. The total training budget is fixed at 100M tokens, with 131,072 tokens per optimizer step, 16 gradient-accumulation steps, 2% warmup, and cosine decay to 20% of the peak learning rate.

The optimizer settings are:

Algorithm 1 High-level Muon update for a matrix parameter

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1: Initialize momentum state  $M_t$ 
2: for  $t = 1, 2, \dots$  do
3:   Compute matrix gradient  $G_t = \nabla_W \mathcal{L}_t(W_{t-1})$ 
4:   Update momentum direction  $M_t \leftarrow \mu M_{t-1} + G_t$ 
5:   Obtain a semi-orthogonalized direction  $O_t$  by applying Newton–Schulz iteration to  $M_t$ 
6:   Update parameters:  $W_t \leftarrow W_{t-1} - \eta_t O_t$ 
7: end for

```

- AdamW baseline: learning rate 6×10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.1.
- Muon family: matrix learning rate 0.023, momentum 0.95, $\beta_2 = 0.9$, weight decay 1.2.

Implemented Muon variant and diagnostics. The implementation is a normalized Muon variant with a Nesterov-style lookahead path. Let G_t denote the matrix gradient and B_t the momentum buffer:

$$B_t = mB_{t-1} + (1 - m)G_t, \quad (4)$$

$$g_{\text{pre},t} = mB_t + (1 - m)G_t, \quad (5)$$

$$g_{\text{post},t} = \mathcal{P}(g_{\text{pre},t}), \quad (6)$$

where $\mathcal{P}$ is the schedule-defined NS map. After orthogonalization, the implementation applies a second-moment normalization on the matrix update magnitude before weight decay and the learning-rate step. The measured geometry is therefore the geometry of the real implemented search direction, not of an idealized standalone matrix-sign iteration.

The study uses three run modes. **train** records validation-loss trajectories, **benchmark** measures end-to-end wall-clock time with synchronization, and **spectral** samples real update matrices for SVD diagnostics. For a sampled matrix X , we use the short-side semi-orthogonality error

$$\varepsilon(X) = \begin{cases} \|X^\top X - I\|_F, & X \text{ tall or square,} \\ \|XX^\top - I\|_F, & X \text{ wide.} \end{cases} \quad (7)$$

This metric is recorded for **buffer_post**, **g_pre**, and **g_post**. We also report validation-loss AUC over tokens and over wall-clock time; these quantities summarize the average validation loss along the whole curve and avoid choosing an arbitrary local slope interval.

3 Main Results

3.1 Coefficient Schedules Change Training Behavior

Table 1 and Figure 1 compare the core five-step schedules under the same model, data, token budget, and learning-rate schedule. The AdamW baseline is separated from the Muon family, which shows that the matrix orthogonalization path changes pretraining behavior substantially in this setting. Within the Muon family, **stable5** is much more conservative than the other schedules. The hybrid **manual.T5.f3.s2** improves over **stable5**, while **fast5** and the two Polar Express schedules form the leading group.

Table 1: Core $T = 5$ training metrics under the fixed 100M-token budget. Lower values are better.

Method	Schedule	Final loss	AUC over tokens
AdamW	adamw	5.3737	5.9677
Stable NS	stable5	4.5273	5.3166
Manual	manual.T5.f3.s2	4.3176	5.0295
Fast Muon	fast5	4.2828	4.9340
Polar Express	pe.T5.l3e-3	4.2873	4.9462
Polar Express	pe.T5.l1e-3	4.2902	4.9338

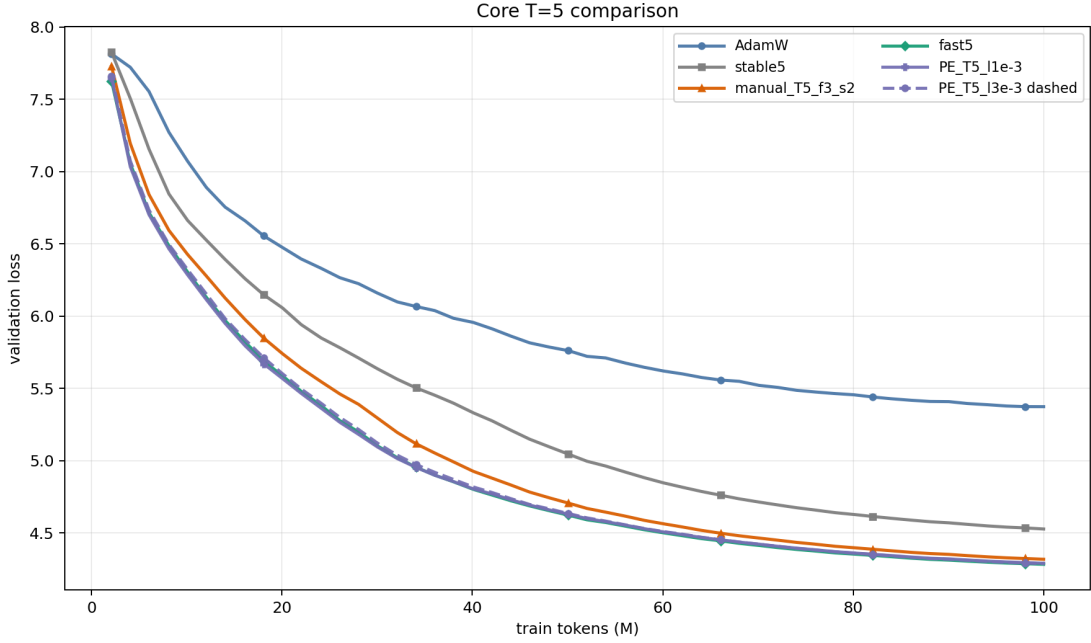


Figure 1: Core $T = 5$ validation-loss curves. The comparison isolates coefficient-sequence effects at fixed depth.

The gap between `stable5` and `fast5` is especially informative. Both use five NS iterations and the same surrounding optimizer path, but their curves differ throughout training. The fast Muon coefficients therefore act as more than a numerical implementation choice: they define a more aggressive singular-value transformation that is useful in this pretraining regime.

3.2 Spectral Maps Explain the Geometry

The training comparison can be interpreted through the composed scalar map $p_T(\sigma)$. For small singular values, it is useful to record both the mapped value and the amplification ratio

$$\text{gain}(\sigma) = \frac{p_T(\sigma)}{\sigma}. \quad (8)$$

The ratio alone can look large when σ is very small, so Table 2 reports it together with $p_T(10^{-5})$. For example, `pe_T9_l3e-5` maps 10^{-5} to about 0.713, which corresponds to a gain of about 7.1×10^4 .

Table 2: Singular-value amplification and post-orthogonalization geometry.

Schedule	$p_T(10^{-5})$	Gain at 10^{-5}	Overall g_post	Attention g_post
<code>stable5</code>	3.20×10^{-4}	32	0.9006	—
<code>fast5</code>	4.85×10^{-3}	485	0.6444	0.5146
<code>manual_T9_f4_s5</code>	4.50×10^{-2}	4502	0.4431	0.1776
<code>pe_T9_l3e-5</code>	7.13×10^{-1}	71309	0.3781	0.0481

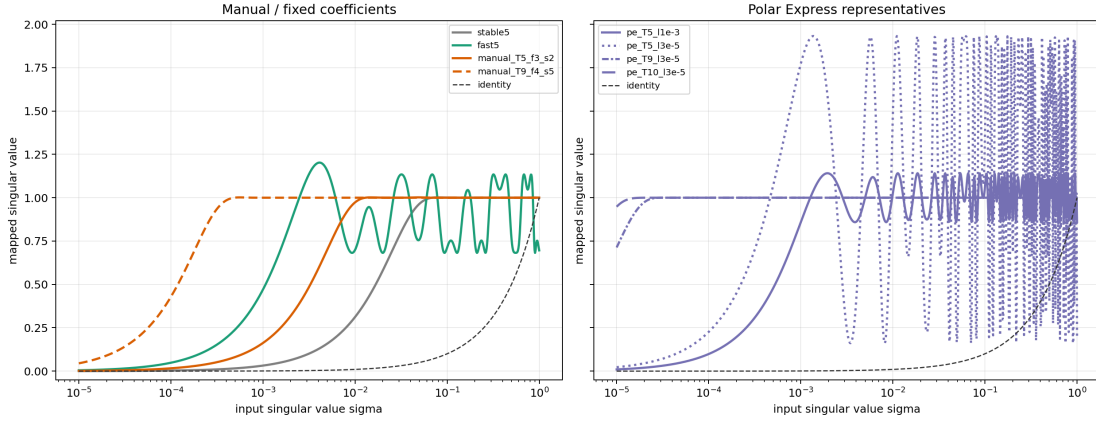


Figure 2: Composed singular-value maps. The schedules differ most clearly in how strongly they amplify small singular values and how smoothly they approach the polar-factor target.

Figure 2 shows the same phenomenon across the full interval. **stable5** is smooth and conservative. **fast5** amplifies small singular values much more strongly. Deeper manual and Polar Express schedules can push small singular values even closer to the polar target, but the maps also become less monotone and more sensitive to the assumed spectral range. This gives a mathematical explanation for why depth and PE lower bound are meaningful algorithmic choices rather than unrelated hyperparameters.

The SVD diagnostics confirm that these scalar maps affect the real optimizer state. Figure 3 reports semi-orthogonality error for **buffer_post**, **g_pre**, and **g_post**. All Muon schedules reduce error after the NS map, but the reduction is schedule-dependent. In particular, **manual_T9_f4_s5** and **pe_T9_13e-5** produce substantially smaller **g_post** error than **fast5**. The module split also shows that the largest schedule-dependent separation appears in attention projection matrices, while MLP matrices are less separated.

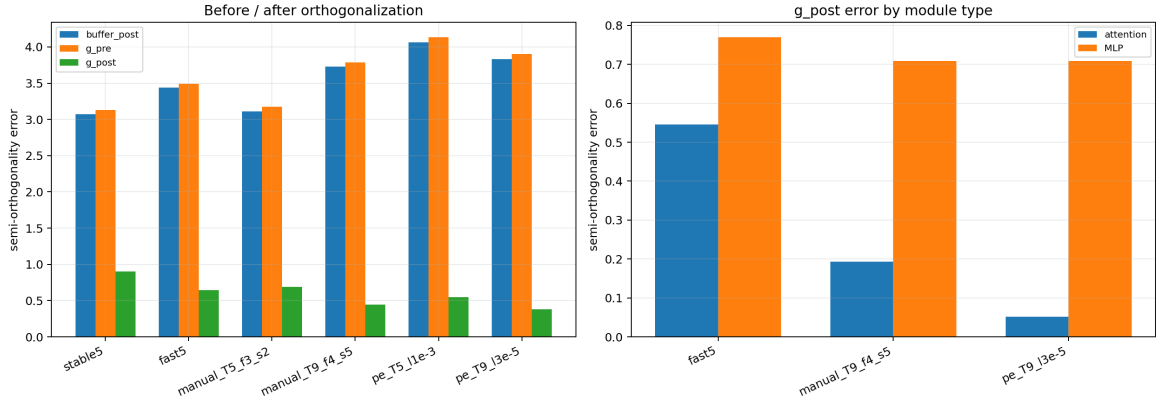


Figure 3: Spectral diagnostics from sampled update matrices. Left: error before and after the NS map. Right: module-level **g_post** error for representative schedules.

3.3 Geometry Only Partially Transfers to Loss and Cost

The spectral results show a clear ordering in update geometry, but the validation-loss comparison is more compressed. To check whether the leading schedules remain separated across random seeds, we repeated **fast5**, **manual_T9_f4_s5**, and **pe_T9_13e-5** with seeds 0, 1, and 2. Table 3 and Figure 4 show that the mean final losses differ by only about 0.01. In this 100M-token setting, **fast5** is the strongest practical baseline, while the geometrically cleaner $T = 9$ schedules do not achieve a reliable validation-loss advantage.

Table 3: Multi-seed final validation loss for representative schedules.

Schedule	Mean final loss	Std. over seeds
fast5	4.2892	0.0056
manual.T9.f4.s5	4.2962	0.0038
pe.T9.13e-5	4.2998	0.0044

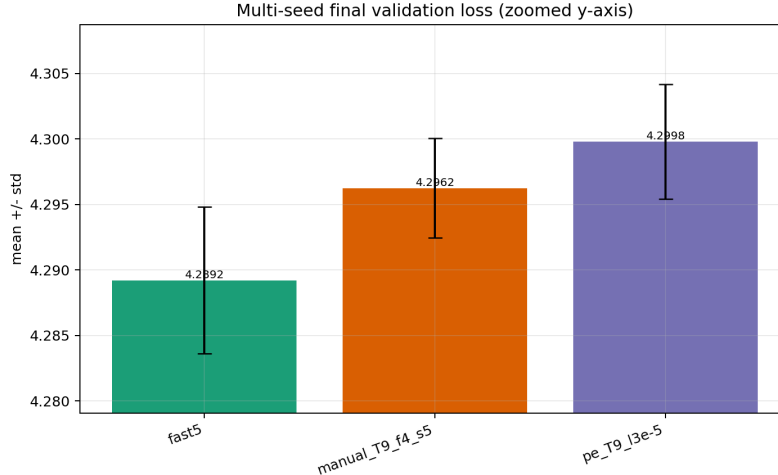


Figure 4: Final validation loss over seeds 0/1/2. The leading schedules are close compared with the AdamW–Muon gap.

The cost comparison gives a similar practical message. At fixed $T = 5$, all Muon schedules finish in about 1095–1099 seconds, so changing scalar coefficients has little effect on runtime once the number of NS steps is fixed. AdamW finishes in 1039.73 seconds because it avoids the matrix orthogonalization path. Deeper schedules are more expensive: **manual.T9.f4.s5** takes 1123.53 seconds and **pe.T9.13e-5** takes 1119.42 seconds. Thus the main tradeoff is not between different five-step coefficient sequences in wall-clock cost, but between stronger geometric correction and the validation-loss benefit obtained from it.

3.4 Schedule-Internal Checks

The internal checks refine the picture without changing the main interpretation. In the manual family, increasing depth from **manual.T5.f3.s2** to the $T = 7$ –10 range improves final loss from 4.3176 to about 4.29, after which the results lie in a narrow band. The best observed representative point is **manual.T7.f4.s3** at 4.2873, while **manual.T8.f5.s3**, **manual.T9.f3.s6**, **manual.T9.f4.s5**, and **manual.T10.f5.s5** are all close. This suggests that extra iterations help the manual family approach the leading schedules, but depth itself is not a monotone quality knob.

Polar Express shows a related sensitivity to the lower-bound parameter. At fixed $T = 5$, **pe.T5.13e-3**, **pe.T5.11e-3**, **pe.T5.13e-4**, and **pe.T5.13e-5** reach final losses 4.2873, 4.2902, 4.2998, and 4.3125 respectively. Since the lower bound determines the singular-value interval used to construct the PE coefficients, this sensitivity is expected from the spectral-map view. Increasing PE depth can repair some low-bound settings, but it also adds computation and does not overturn the multi-seed comparison with **fast5**.

4 Conclusion

This project compared Muon schedules by treating Newton–Schulz coefficients as spectral maps. The first main finding is that coefficient choice matters even at fixed depth: the stable NS schedule is too conservative for this pretraining setting, while the fast Muon coefficients and well-chosen Polar Express schedules give much better token efficiency. This answers the first part of the project question at the training level and is supported by the fixed $T = 5$ comparison.

The second finding is geometric. The composed maps $p_T(\sigma)$ explain how different schedules amplify small singular values, and the sampled update matrices show the resulting change in semi-orthogonality error. Deeper manual and PE schedules produce cleaner post-orthogonalization matrices, especially in attention layers. This confirms that the coefficient schedule changes the actual update geometry, not only the notation of the optimizer.

The third finding concerns the connection between geometry and optimization quality. The cleaner updates produced by deeper manual and PE schedules do not reliably reduce final validation loss relative to **fast5** at the current 100M-token scale. The relationship between semi-orthogonality and language-model loss is therefore not monotone in these experiments.

A schedule can make the update closer to a polar factor while still failing to improve the full stochastic training trajectory.

The study is limited to one model scale, one token budget, and a small multi-seed set for representative schedules. Larger runs would be needed to determine whether the attention-layer spectral separation persists at scale. On the mathematical side, the natural next step is to analyze the fixed points, contraction regions, and lower-bound assumptions of the composed quintic maps, then relate those properties to the measured training trajectories.

References

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